Nagar YuwakShikshanSanstha's

YeshwantraoChavan College of Engineering

(An Autonomous Institution affiliated to RashtrasantTukadojiMaharaj Nagpur University)

(Accredited 'A' Grade by NAAC)

Hingna Road, Wanadongri, Nagpur - 441 110

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Form No YI-04

Department of Electronics Engineering

STUDENT'S DIARY/ LOG

Your Name: Khushi Narendra Atara

Your Institute: Yeshwantrao Chavan College of Engineering Nagpur

Company where internships are carried out : Cognizant Technology Solutions Pvt. Ltd

Period: From: 31/01/25 To: 13/05/26

Week: 14		DATE: From	04/05/26 to 08/05/26
Dept./Division	Programmer Analyst Trainee	Name of Project	Training and Internship Program
Name of HOD/ Supervisor With e-mail id		Name: Ashly M. Benny	
		Email Id: ashly.bennym@cognizant.com	
Main points of the week		Stage 2 – REST Assured Advanced, Docker & Containers, GenAI Hands-On & Hackathon Delivery (Days 58–62)	

• Practiced REST Assured authentication: Basic Authentication, Digest Authentication, API Key, Bearer Token, and OAuth 2.0 in both Postman and REST Assured framework.

• Completed POJO-based serialization and deserialization in REST Assured framework for structured request/response handling.

• Studied payload handling in REST Assured: sending JSON file, XML file, form-data parameters, and multipart file upload/download.

• Completed REST Assured hands-on: university details for USA/India/Canada, population validation, authentication APIs, and dynamic data generation using Faker (Day 61).

• Created Cucumber BDD Framework supporting both API and Web automation; integrated REST Assured with TestNG/Maven and Jenkins Freestyle project.

• Studied Docker and Containers: Introduction to Docker, Docker images and containers, running Selenium tests using Docker containers, setting up Dockerized Selenium Grid for consistent environments (Day 62).

• Completed GenAI Comprehensive Hands-On: Chat Endpoint Basics, Tokenization and Context Limits, Prompt Optimization, Generate Edge Cases, AI-Driven BDD Generation, Dynamic Locator Generation, and AI-Augmented Framework.

• Submitted all remaining Integrated Project Deliverables: API Smoke Suite Postman Collection, REST Assured Automation Module, Comprehensive Automation Strategy and Consolidated Execution Reports (due Day 61).

Signature and stamp of Industry Supervisor

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Fortnightly Progress Report

(To be mailed every alternate Monday to Project Guide, designated person(s) of the company, your HoD and the concerned Dean from your institute)

1. Company Overview:

Cognizant Technology Solutions is a leading global IT services and consulting company headquartered in the USA, with a major delivery presence across India. It specializes in digital transformation, cloud computing, AI, automation, software engineering, and quality assurance. Cognizant's QEA (Quality Engineering and Assurance) division focuses on intelligent and automated quality assurance, modernisation assurance, and experience assurance for enterprise clients across healthcare, finance, and technology sectors.

2. Who did you interact with in this period? Which departments did you visit/work in?

During Week 14, I interacted with Cognizant Academy trainers during Master Sessions on advanced REST Assured topics and Docker containerization (Days 58–62). On Day 62, I delivered the completed Hackathon project to the BU SME for review and incorporated feedback. All interactions were within the Training division. This was the final week of technical learning content before the Final Evaluation week.

3. List down (maximum) three activities carried out by you in this period. Describe each one briefly.

Activity 1 – Advanced REST Assured (Days 58–61): Completed all remaining REST Assured chapters covering authentication methods (Basic, Digest, API Key, Bearer Token, OAuth 2.0), POJO serialization/deserialization, Request/Response Specification builders, payload handling (JSON, XML, form-data, multipart), and parallel test execution with TestNG. Also created a Cucumber BDD framework supporting both API and web automation and integrated it with Jenkins.

Activity 2 – Docker & Containers (Day 62): Attended the Master Session on Docker fundamentals including Docker images and containers, running Selenium tests inside Docker containers, and setting up a Dockerized Selenium Grid for consistent cross-environment test execution. Also completed the comprehensive GenAI Hands-On exercises covering AI-driven BDD generation, dynamic locator generation, and AI-Augmented framework building using GitHub Copilot.

Activity 3 – Hackathon Project Final Delivery: Delivered the completed group Hackathon project to the BU SME on Day 62 for evaluation. The project demonstrated the complete automation framework including Selenium WebDriver, POM, TestNG, Cucumber BDD, REST Assured, Jenkins CI/CD, and Docker integration. All integrated project deliverables including the API Smoke Suite, REST Assured Automation Module, and Consolidated Execution Reports were also submitted.

4. What problems did you face and what did you do to address them? What was the learning that resulted from each of them?

Faced challenges in setting up the Dockerized Selenium Grid with the correct node configurations for different browser versions. Referred to BU Chapter 17 Docker content and the Udemy Section 13 Docker material for detailed configuration steps. This experience highlighted the importance of containerization in ensuring consistent

test execution environments across development, testing, and CI/CD pipeline stages.

5. What process(s)/system(s) did you familiarize yourself with/study as part of your work or out of your interest?

Familiarized myself with Docker containerization for creating isolated, reproducible test execution environments using Dockerized Selenium Grid. Also completed the comprehensive GenAI Hands-On module covering prompt engineering techniques with GitHub Copilot for generating automation test code, BDD feature files, and test edge cases. Studied the complete REST Assured framework covering serialization, authentication, and parallel API test execution integrated with Maven and Jenkins CI/CD.

6. If you have seen anything where an improvement is possible/required, please mention/describe that here.

A guided walkthrough session showing how to containerize a complete Selenium automation framework – from writing the Dockerfile to running the test suite on a Dockerized Grid and publishing reports – would be very beneficial. Currently the Docker module is covered at a conceptual level and more hands-on time with Docker Compose for multi-container Selenium Grid setups would greatly strengthen practical understanding before the Final Evaluation.

7. What did you find novel/exciting/valuable during this period?

The GenAI Comprehensive Hands-On using GitHub Copilot was the most exciting experience of this week. Watching GitHub Copilot generate complete Selenium test scripts, BDD feature files, and dynamic XPath locators from plain English prompts demonstrated how AI is transforming the future of test automation. This gives a strong sense of how QA engineers at Cognizant will increasingly use AI assistance to accelerate automation development and improve test coverage with minimal manual effort.

8. What according to you, can be of use to your institute and students and in what ways?

Introducing Docker as part of the academic DevOps or cloud computing curriculum would prepare students for modern software delivery practices. Understanding how to containerize applications and test environments is a highly sought-after skill in the industry. Additionally, introducing GitHub Copilot or similar AI coding assistants in academic programming labs would expose students to AI-assisted development early, making them more productive and industry-ready from day one of their careers.

9. Have you been able to complete everything that you had planned last fortnight? If not, what have you not completed and why?

Yes, all planned activities for Week 14 were completed. The advanced REST Assured hands-on exercises (Chapters 7–18 API Fundamentals), Docker and Selenium Grid containerization session, comprehensive GenAI hands-on exercises, Hackathon project final delivery to BU SME, and all remaining Integrated Project deliverables including API Smoke Suite, REST Assured Automation Module, and Consolidated Execution Reports were all completed within the scheduled program timeline.

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Khushi N. Atara
Name and Signature of Student.

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